

Kelechi G. Kalu

PhD Candidate
Elmore Family School of Electrical and Computer Engineering
Purdue University
West Lafayette, IN 47906

kalu@purdue.edu
765-767-3626
<https://kalu-kelechi-gabriel.github.io/>

RESEARCH THEME

My research examines how developers, security practitioners, and organizations adopt and use security mechanisms in real software and AI systems. I focus on software supply-chain security, especially signing, provenance, verification, transparency, and trust in open-source ecosystems and developer workflows. Using interviews, usability studies, surveys, and large-scale measurement, I study where security tools succeed, where they create friction, and how their design can better support real users. My broader goal is to make security mechanisms technically sound, usable, deployable, and trustworthy in practice.

EDUCATION

- Ph.D, Electrical and Computer Engineering

Purdue University, West Lafayette, IN

2023-2027
- B.Eng. Electronic and Computer Engineering

Nnamdi Azikiwe University, Awka, Nigeria

PROFESSIONAL EXPERIENCE

- Graduate Research Assistant

ECE@Purdue University — Advised by Dr. James C. Davis

2023–present

- Led qualitative, quantitative, and longitudinal studies of software signing tools, focusing on the adoption, usability, and effectiveness of software signing in developer workflows. This work resulted in publications at USENIX Security 2025 and USENIX Security 2026, as well as a technical report under submission.
 - Investigated provenance consistency across software publishing pipelines to identify gaps in artifact traceability and strengthen software supply-chain assurance.
 - Conducted an NSF-supported scoping review on research-software supply-chain security, focusing on terminology, threats, controls, and ecosystem-specific risks.
 - Collaborated on empirical studies of coding agents and pre-trained model recommendation, extending my research agenda toward AI-enabled software engineering workflows.
 - Published 3 top-tier conference papers, 4 workshop/short papers, 4 technical reports, and 3 posters; contributed to an awarded NSF grant and sponsor-facing research outputs (like Google).
 - Mentored Ph.D., master’s, and undergraduate students on empirical security research, including study design, data analysis, and technical writing.
- Research Intern (MSR-Cryptography)

Microsoft Research, Redmond, WA — Mentored by Dr. Kim Laine and Dr. Betul Durak

Jun.–Aug. 2024

- Researched conceptual designs and models for actor reputation to improve the security of the software supply chain in the GitHub ecosystem.
 - Designed a data-collection pipeline for GitHub maintainers and projects; piloted on Azure Repos data with 10,000+ user entities.
 - Designed and experimented with trust metrics for an actor reputation system.
 - Published results in a peer-reviewed technical report (*ICSE-JAWs’26*).
- Software Engineer

Enugu Electricity Distribution Company, Enugu, Nigeria

Oct. 2019–Dec. 2022

- Designed, developed, and scaled business-critical applications serving approximately 2 million customer accounts in Southeast Nigeria.
 - Key systems: ICT & Administration Support web app; Arrears One-Time Settlement (OTS) platform; Customer Feedback web app; Attendance Management System; Meter Installation application.
- Software Engineer

Afrivelle Technologies, Enugu, Nigeria

Jun. 2019–Sep. 2019

- Developed the administrative backend and designed the database for the real-estate website neneakintanrealestate.com (now rebranded as Oakville Living with Nene).
- Software Engineer

Playjoor Technologies, Enugu, Nigeria

May 2018–Apr. 2019

- Deployed a Lichess (chess) server for the playjoor.com web application and managed AWS EC2 test-server administration, serving a few thousand customers.

Research Intern

May 2016–Oct. 2016

Electronics Development Institute, Awka, Nigeria

- Researched fuzzy-logic control via a systematic literature review and MATLAB simulations, and developed implementations of fuzzy controllers on PIC microcontrollers.

REFEREED CONFERENCE PUBLICATIONS

- [1] **Kalu**, Okorafor, Singla, Chen, Torres-Arias, Davis. *Why Johnny Adopts Identity-Based Software Signing: A Usability Case Study of Sigstore*. Proceedings of the 35th USENIX Security Symposium. 20 pages. (**USENIX Security’26**).
- [2] **Kalu**, Singla, Okafor, Torres-Arias, Davis. *An Industry Interview Study of Software Signing for Supply Chain Security*. Proceedings of the 34th USENIX Security Symposium. 17.1% acceptance rate (407/2,385). 18 pages. (**USENIX Security’25**).
- [3] Schorlemmer, **Kalu**, Chigges, Ko, Ishgair, Bagchi, Torres-Arias, Davis. *Signing in Four Public Software Package Registries: Quantity, Quality, and Influencing Factors*. Proceedings of the 45th IEEE Symposium on Security and Privacy. 18% acceptance rate (261/1463). 16 pages. (**IEEE S&P 2024**).

OTHER REFEREED WORKS: WORKSHOP, VISIONS, POSITION, MAGAZINE ARTICLES

- [1] **Kalu**, Okorafor, Durak, Laine, Moreno, Torres-Arias, Davis. *ARMS: A Vision for Actor Reputation Metric Systems in the Open-Source Software Supply Chain*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering). Recommended for journal extension; 30/79 papers. 9 pages. (**ICSE-JAWs’26**).
- [2] **Kalu**, Rattan, Schorlemmer, Thiruvathukal, Carver, Davis. *Operationalizing Research Software for Supply Chain Security*. The First Journal Ahead Workshop (co-located with the 2026 International Conference on Software Engineering). 5 pages. (**ICSE-JAWs’26**).
- [3] Peng, Zhong, Méndez, **Kalu**, Davis. *How Do Agents Perform Code Optimization? An Empirical Study*. Proceedings of the 23rd International Mining Software Repositories Conference — Mining Challenge track. 53% acceptance rate (62/116). 5 pages. **LLM/Agentic systems work (MSR-Mining’26)**.
- [4] Patil, Jiang, Peng, Lugo, **Kalu**, LeBlanc, Smith, Heo, Aou, Davis. *Recommending Pre-Trained Models for IoT Devices*. Proceedings of the 7th International Workshop on Software Engineering Research & Practices for the Internet of Things. 26% acceptance rate (25/97). 5 Pages. (**SERP4IoT’25**).
- [5] Schorlemmer, Burmane, **Kalu**, Torres-Arias, Davis. *Establishing Provenance Before Coding: Traditional and Next-Gen Signing*. IEEE Security & Privacy Magazine, special issue “Secure Software Before Coding”, 2025. 8 Pages. (**IEEE S&P Magazine’25**).
- [6] **Kalu**, Schorlemmer, Chen, Robinson, Kocinaire, Davis. *Reflecting on the Use of the Policy-Process-Product Theory in Empirical Software Engineering*. Proceedings of the 31st ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering- Ideas Visions, and Reflections track. 48% acceptance rate (13/27). 5 Pages. (**ESEC/FSE-IVR’23**).
- [7] Singla, Anandayuvraj, **Kalu**, Schorlemmer, Davis. *An Empirical Analysis of Large Language Models to Analyze Software Supply Chain Security Failures*. ACM Workshop on Software Supply Chain Offensive Research and Ecosystem Defenses. 67% acceptance rate (14/21). 11 Pages. **LLM/Agentic systems work (SCORED’23)**.

TECHNICAL REPORTS

- [1] **Kalu**, Tran, Torres-Arias,, Jeong, and Davis. *A Longitudinal Study of Usability in Identity-Based Software Signing*. <https://arxiv.org/abs/2603.17133>. 2026.
- [2] **Kalu**, Davis. *Why Software Signing (Still) Matters: Trust Boundaries in the Software Supply Chain*. <https://arxiv.org/pdf/2510.04964>. 2025.
- [3] Davis, Chen, Peng, Amusuo, and **Kalu**. *A Guide to Stakeholder Analysis for Cybersecurity Researchers..* <https://arxiv.org/pdf/2508.14796>. 2025.

POSTERS

- [1] **Kalu**, Tran, Jeong, Torres-Arias, and Davis. *A Longitudinal Study of Usability in Identity-Based Software Signing*. Center for Education and Research in Information Assurance and Security (CERIAS) Purdue Poster session 2026 (**CERIAS’26**).
- [2] **Kalu**, Davis. *Investigating Software Provenance Consistency in the Open Source Publishing Pipeline*. Poster Session of the 34th USENIX Security Symposium (**USENIX Security–Poster’25**).
- [3] **Kalu**, Torres-Arias, and Davis. *Software Signing: Practical Adoption, Challenges, and Tooling Usability*. Center for Education and Research in Information Assurance and Security (CERIAS) Purdue Poster session 2025 (**CERIAS’25**). **Best Poster Award – 1st Position**.
- [4] **Kalu**, and Davis. *Navigating Software Supply Chain Risks: Practitioner Perspectives on Software Signing*. Center for Education and Research in Information Assurance and Security (CERIAS) Purdue Poster session 2025 (**CERIAS’24**).

TECHNICAL SKILLS

Research Methods: Semi-structured interviews, usability studies, survey design, thematic analysis, framework analysis, empirical measurement

Programming: Python, JavaScript, PHP, SQL, C

Web & Databases: React.js, Node.js, Laravel, CodeIgniter, MySQL, PostgreSQL, MongoDB

Tools & Platforms: Git, Docker, AWS EC2, MATLAB, Google Data Studio, Excel.

INVITED TALKS

Software Signing in Practice: Lessons from Adoption and Usability Toward Broader Supply Chain Trust 2026
The Center for Education and Research in Information Assurance and Security (CERIAS), Purdue University: <https://tinyurl.com/kelechi-kalu-CERIAS>

AWARDS AND RECOGNITION

CERIAS Research Poster Competition **CERIAS’25** – 1st Position 2025

USENIX Security Student Grant **USENIX’25** 2025

ACM SIGSOFT CAPS Travel Grant, **ESEC/FSE’23** 2023

NSF Student Travel Award, **ESEC/FSE’23** 2023

Agbami/Chevron Medical and Engineering Scholarship Award – Nnamdi Azikiwe University 2014-2017

SERVICES

Artifact Evaluation Committee: USENIX Security Symposium 2026; Privacy Enhancing Technologies Symposium (PETS) 2026.

Program Committee Service: Junior Program Committee Member, MSR 2026 and TechDebt 2026; Shadow Program Committee Member, ICSE 2026; Program Committee Member, SCORED 2025.

Reviewing: Sub-reviewer for ICSE, USENIX Security, FSE, ISSTA, IEEE S&P, ESEC/FSE, and ASE, 2023–2026.

Academic and Community Service: Judge, Purdue SURF Poster Competition, 2025; Student Volunteer, USENIX Security 2025 and ESEC/FSE 2023; Panelist, Purdue ECE Spring Graduate Open House, 2025.

GRANT WRITING

[1] **NSF #2537308: Collaborative Research: Planning: CROSS: Building a Community aROund Securing the Research Software Supply Chain**
PI: James C. Davis
US National Science Foundation
2025–2027. \$105,113.

PROFESSIONAL MEMBERSHIPS

Member, Institute of Electrical and Electronics Engineers (IEEE)

Member, Association for Computing Machinery (ACM)